

Digital Farming Assistant

Purpose

The goal of the Digital Farming Assistant is to empower farmers through a digital platform that uses data-driven technologies to enhance agricultural productivity and decision-making. The reason why we think the problem is important is because of general limitations found in farming apps. Such as:

- **Digital Literacy Barriers:** Many marginal farmers struggle with app navigation, reading complex text prompts, or completing multi-step digital forms.
- **Lack of Hyper-Local Precision:** Soil profiles and micro-climates vary widely across small regions; generalized app data may not match specific field conditions.
- **Lack of connectivity:** Many of them do not have any kind of forum where people can share their concerns and get feedback on it.
- **Fragmented Ecosystems:** Separate apps handle crops, livestock, government sales, and weather, meaning users must switch between multiple platforms instead of using a single unified assistant.
- **Crop Disease Treatment & Prevention Advice**
- **Location-Based Weather Forecast**
- **Agricultural Weather Alerts**
- **Diagnosis History**

Objectives of this system includes:

- ❖ Providing AI/ML-based crop health detection and feedback.
- ❖ Offering real-time weather forecasts to aid crop planning.
- ❖ Maintaining a crop library with cultivation guidelines.
- ❖ Creating a community forum for farmers to share experiences and solutions.
- ❖ Simplifying user access with a minimal login system (name and phone number only).

Inception:

The Inception phase identifies the stakeholders, their needs, and their expectations from the proposed Digital Farming Assistant. The main stakeholders include farmers, agricultural experts, administrators, project sponsors, the development team, weather data providers, and government agriculture officers.

Farmers need easy access to crop diagnosis, farming information, weather updates, AI assistance, and expert support. Experts and government officers focus on accurate and verified agricultural information, while administrators focus on security, content management, and quality assurance. The development team and project sponsor focus on feasibility, project scope, quality, and successful completion.

This phase also identifies common and conflicting requirements among stakeholders so that they can be prioritized and combined into the final system requirements.

2.1 Identifying Stakeholders

ID	Name/Role	Organization	Position	Why They Are a Stakeholder	Expectations	Type of Interactions
S 1	Project Sponsor	East West University	Final academic authority, approves scope	Sponsors the project.	Successful implementation, academic quality and achievement of project objectives.	Reviews reports, approves scope.
S 2	Project Manager	Project Team	Leads execution and coordination	Lead the Development Team.	Timely completion, proper coordination and successful delivery.	Uses the system for management monitoring and coordination.
S 3	HR Consultant (Dr. Rajesh Chakraborty)	Shere Bangla Agricultural University	Advisory	Provides professional advice.	A practical and user-friendly system that addresses stakeholder needs.	Provides feedback, reviews requirements, and gives recommendations.
S 4	Development Team	EWU Students	Designs and develops system	Develops the entire system.	Clear requirements, technical feasibility, and successful system operation.	Directly develops, tests, updates, and maintains the system.
S 5	Farmer Representative	Farming Community	Represent end users	Represents the primary end users of the system.	Easy access to useful farming information and communication with other farmers.	Uses the app, provides feedback, and communicates through the farmer community.
S 6	Agricultural Expert	Agriculture Sector	Provides crop guidance knowledge	Provides reliable agricultural knowledge for the system	Accurate and useful agricultural information reaching farmers.	Provides/updates guidance and responds to agricultural queries through the platform.
S 7	Weather Data Provider	Meteorological Dept	Supplies forecast data	Provides weather information needed for	Accurate and timely presentation of weather information.	Provides weather data through data integration/API or regular updates.

				farming decisions.		
S 8	Government Agriculture Officer	Govt. of Bangladesh	Regulatory Oversight, Quality Assurance	Represents the government agricultural support.	A reliable system that supports farmers and complements agricultural services.	Reviews information, provides guidance, and may interact with farmers through the platform.

2.2 Identifying multiple viewpoints of the stakeholders

1. Project Sponsor

Project sponsors usually want a complete and functional Digital Farming Assistant that addresses a real agricultural problem. Project sponsor might consider User registration/login, Bengali/English support, auto location detection, AI crop health diagnosis, treatment/prevention advice, diagnosis history, location based weather forecasts, agricultural weather alerts, searchable crop library, offline crop info, farmer community forum, image sharing/comments/likes, notifications, AI agricultural chatbot, user management, content moderation, activity logs, regulatory oversight/quality assurance as important feature. While their main concern would be data protection of the user.

2. Project Manager

Project managers want a system with clearly defined and manageable requirements and successful completion of all agreed functionality. They consider user registration/login, role-based access, user management, AI crop diagnosis, diagnosis history, AI agricultural chatbot, weather/agricultural alerts, crop information library, offline functionality, community forum, notifications, admin dashboard, content moderation, security/activity logs, input verification/quality assurance important while having limited development time and resources.

3. HR Consultant

They want a practical agricultural assistance system that addresses real problems faced by farmers.

They consider AI crop health diagnosis, crop disease treatment/prevention advice, searchable crop library, AI agricultural chatbot, diagnosis history, farmer community forum, agricultural expert involvement, Bengali language support, input verification, regulatory oversight/quality assurance, weather/agricultural alerts as important features.

What constraints or concerns do they have? **Funny pore dekho nicher tuku**

- AI-generated agricultural recommendations may not always be completely accurate.
- Treatment recommendations should be based on reliable agricultural knowledge.
- The system should clearly distinguish AI-generated guidance from verified expert information.
- Fertilizer recommendations should consider factors such as crop, field area, and available agricultural information.
- Harmful or incorrect agricultural advice should be prevented from being presented as verified information.
- Agricultural content should be reviewed or verified by appropriate knowledgeable stakeholders.

4. Development Team

This stakeholder wants clear and complete requirements and well-defined functionality for each type of user.

They consider important features like : Registration/login, Bengali/English interface, location detection, AI crop diagnosis, treatment/prevention advice, diagnosis history, weather forecasting/alerts, crop information library, offline info/diagnosis queue, community forum, image sharing/comments/likes, notifications, AI chatbot, admin dashboard, user management, content moderation, security/activity logs, regulatory/input verification functions.

They have constraints such as:

- Limited project time and resources.
- AI diagnosis and chatbot responses may require external AI services.

5. Farmer Representative

This stakeholder wants a simple application that can help farmers with everyday agricultural problems.

What features do they consider Bengali language support, simple registration/login, automatic location detection, AI-based crop diagnosis, crop disease treatment/prevention advice, diagnosis history, location-based weather forecast, agricultural weather alerts, crop information library, offline crop information, offline diagnosis queue, farmer community forum, image sharing, comments/likes, notifications, AI agricultural chatbot, fertilizer/farming-related calculations/recommendations as main features.

They have many constraints to address such as:

- Some farmers may have limited technical knowledge.
- Internet connectivity may be unreliable in rural areas.
- The interface should be simple and easy to understand.

6. Agricultural Expert

This stakeholder wants a reliable platform for providing agricultural guidance and access to relevant farmer questions and crop information from the system.

They consider Expert access, crop diagnosis information, farmer questions, community forum, AI chatbot information requiring expert verification, crop information library, treatment/prevention recommendations, regulatory oversight, quality assurance, input verification, notifications, activity records where appropriate. important.

There constraint is:

- Agricultural information should be verified before being presented as authoritative.

7. Weather Data Provider

This stakeholder wants Weather information to be presented accurately and appropriately and Timely delivery of relevant weather information to farmers from the system.

They consider Automatic location detection, location-based/current weather forecasts, agricultural weather alerts, heavy rainfall warnings, other relevant weather notifications, notification delivery as important features.

As weather forecasts can change over time and it could have uncertainty, it is a concern for them.

8. Government Agriculture Officer

This stakeholder wants a reliable agricultural support platform and accurate agricultural information reaching farmers from the system.

Crop information library, AI crop health diagnosis, treatment/prevention advice, AI agricultural chatbot, fertilizer/farming recommendations, agricultural weather alerts, expert consultation, input verification, quality assurance, regulatory oversight, content moderation, security/activity logs, Bengali language support are some they do consider important.

Constraints :

- Agricultural recommendations should not contradict appropriate official guidance.
- Important agricultural information should be verified before being treated as authoritative.

2.3 Working towards Collaboration

Common Requirements

Things nearly every stakeholder independently asked for:

- User registration/login with Bengali language support
- AI-based crop health diagnosis with treatment/prevention advice
- Diagnosis history
- Searchable crop information library
- Location-based weather forecasts and agricultural alerts
- AI agricultural chatbot
- Farmer community forum
- Notifications
- Content moderation

- Input verification / quality assurance for agricultural information
- Security and activity logging
- Clear distinction between AI-generated and expert verified information

Conflicting Requirements

- **Speed/availability of AI advice** (Farmer, Sponsor) **vs. mandatory expert verification before trust** (Expert, Govt Officer, HR Consultant). Verification adds delay that undermines "quick assistance" farmers want.
- **Full featured, comprehensive system** (Sponsor, HR Consultant, Govt Officer) **vs. limited time/resources and manageable scope** (PM, Dev Team). Feature wish-list exceeds academic project bandwidth.
- **Open, freely shared community forum** (Farmer) **vs. strict content moderation and regulatory oversight** (Govt Officer, Sponsor) .
- **Simple, low tech interface for low literacy users** (Farmer) **vs. robust security** (PM, Dev Team, Sponsor).
- **Automatic location detection for convenience** (Sponsor, Weather Provider) **vs. user privacy** (Farmer, Govt Officer).
- **Presenting weather data with full uncertainty** (Weather Provider) **vs. simple, non-technical answers** (Farmer) .
- **Offline first operation** (Farmer, Dev Team) **vs. real-time sync integrity and up to date verified content** (Sponsor, Expert) .

Final Requirements

- Bengali first UI with English toggle; simple, low literacy friendly design; optional SMS/voice fallback for feature-phone users
- AI crop diagnosis with a clearly labeled confidence/uncertainty indicator, routed to expert review when confidence is low
- Two-tier information system: "AI suggested" vs. "Expert verified" content, visually distinguished
- Offline mode limited to previously cached content + a diagnosis request queue that syncs and re-validates once online
- Location detection as opt-in only
- Weather forecasts displayed with timestamp
- Community forum with lightweight moderation

- Role based access (Farmer, Expert, Moderator, Admin) with security logging scoped to protect farmer data privacy
- Registration/login, AI diagnosis + history, weather alerts, crop library, chatbot, basic forum, admin dashboard and offline queue as core features, since these directly address the Expert/Govt accuracy concern
- Deferred items : market price info, mobile payment integration, supplier directory, dialect support noted as valuable but outside academic timeline.

Elicitation

3.1.1 Normal Requirements

Explicitly stated by stakeholders during elicitation which satisfy customers when present, dissatisfy when absent.

- User registration and login
- Bengali and English language support
- Automatic location detection (with permission)
- AI-based crop health diagnosis
- Crop disease treatment and prevention advice
- Diagnosis history
- Location-based weather forecasts
- Agricultural weather alerts
- Searchable crop information library
- Offline crop information and diagnosis queue
- Farmer community forum with image sharing, comments, and likes
- Notifications
- AI chatbot for agricultural questions
- Admin dashboard and user management
- Content moderation
- Security and activity logs

- Regulatory oversight, input verification, and quality assurance
- Role-based access for farmer, expert, moderator, and administrator

3.1.2 Expected Requirements

Fundamental like: rarely stated outright because stakeholders assume them, but their absence causes real dissatisfaction

- Data security and encryption for personal information and uploaded crop images
- System availability/uptime and graceful handling of external service failures (AI service, weather API downtime)
- Reasonable response time for diagnosis and chatbot queries
- Basic input validation and error handling (e.g., invalid image uploads, empty forms)
- Consistent, timestamped data (so users know how current the weather/diagnosis info is)
- Cross-device/screen-size responsiveness for the app UI
- Data backup and recovery for user records and diagnosis history
- Clear permission prompts before accessing location, camera, or storage
- Sync conflict handling when offline queue reconnects
- Basic accessibility (readable font sizes, sufficient contrast) for older or low-literacy farmers
- Audit trail for admin/moderator actions (who approved/rejected what content)
- Session management
- Scalability to handle a growing number of users and records without major redesign

3.1.3 Exciting Features

- Personalized crop calendar with harvest reminders based on location and season
- Gamification of community participation (badges, reputation score for helpful farmers/experts)

4. Scenario-Based Modeling

4.1 Usage Scenarios

Scenario 1: User Registration and Login

Actor: Farmer

Goal: Access the Digital Farming Assistant securely.

The farmer opens the application and creates an account by providing the required information. After registration, the farmer logs in using their credentials and accesses the application's features.

Scenario 2: Crop Health Diagnosis

Actor: Farmer

Goal: Identify a possible crop disease or health problem.

The farmer takes or uploads a photo of an affected crop. The system analyzes the image and provides a possible diagnosis along with treatment and prevention advice. The result is saved in the farmer's diagnosis history.

Scenario 3: Weather Forecast and Alerts

Actor: Farmer

Goal: Obtain weather information relevant to farming.

The application detects the farmer's location with permission and provides the relevant weather forecast. If important weather conditions such as heavy rainfall are detected, the system sends an agricultural weather alert.

Scenario 4: AI Agricultural Chatbox

Actor: Farmer

Goal: Get quick answers to agricultural questions.

The farmer enters a question, such as the required fertilizer quantity for a specific field area. The AI chatbox processes the question and provides relevant agricultural guidance.

Scenario 5: Crop Information Search

Actor: Farmer

Goal: Learn about a specific crop.

The farmer searches for a crop in the information library. The system displays relevant information about cultivation, fertilizer, pests, diseases, and farming practices.

Scenario 6: Farmer Community

Actor: Farmer

Goal: Share information and discuss farming problems.

The farmer creates a community post and may attach a crop image. Other farmers can view the post, comment, and like it. The farmer receives notifications about relevant interactions.

Scenario 7: Expert Consultation

Actor: Farmer / Agricultural Expert

Goal: Receive professional agricultural guidance.

The farmer submits a question or crop-related problem to an agricultural expert. The expert reviews the information and provides appropriate recommendations through the platform.

Scenario 8: Offline Diagnosis Queue

Actor: Farmer

Goal: Use important features when there is no internet connection.

The farmer submits a crop diagnosis request while offline. The application stores the request and processes it when an internet connection becomes available.

Scenario 9: Administration and Content Moderation

Actor: Administrator

Goal: Manage and maintain the platform.

The administrator logs into the dashboard to manage users, agricultural content, experts, notifications, and reported community posts. The administrator can review and moderate inappropriate content.

Scenario 10: Agricultural Information Verification

Actor: Agricultural Expert / Government Agriculture Officer

Goal: Maintain the quality and reliability of agricultural information.

The authorized stakeholder reviews agricultural information, recommendations, or important inputs. The information can be verified, updated, or flagged when necessary before being treated as verified guidance in the system.

